

Shortened Abstract — When a Model Is Not Retained

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Short abstract (≈ 230 words), condensed from carrier paperF1_retention_framework_v30_humanized (stamp se30-c1375ed), 18 September 2026. All claims below are drawn from the frozen values of the full abstract; nothing new is asserted.

Process-based forecasts are elaborated by default, yet the added structure is rarely required to beat a naive baseline under a decision rule fixed before scoring—and where it fails, the failure is uninterpretable without knowing the instrument’s power. This article proposes a three-part minimum reporting standard: an explicit, executable retention algorithm; an information-set audit separating inputs available at the forecast origin from those supplied later; and a mandatory operating-characteristic simulation quantifying power and specificity under known truth. Demonstrated on three scored objects from two domains—Northern cod (NAFO 2J3KL, Specifications 1983–2015 and 1954–2024) and Edwards Aquifer index well J-17 (1934–2023)—the retained set is empty everywhere. On cod, no structural candidate nears the 5% tie band (persistence 98.05 kt versus 114.80 kt at one year; closest margin +9.01% at five years), and every model misses the 1991–1995 collapse (694–819 kt versus 670 kt). On Edwards, a climatological-flux map beats persistence (12.283 versus 13.230 ft; 17.445 versus 21.106 ft at five years) but is declined on pre-specified class grounds, its comparator margin of 4.33% falling inside the band; an oracle with realized future fluxes bounds one-year error at 7.547 ft (–42.96%). In simulation, the rule is specific (specificity 0.978) and powerful for autonomous dynamics (0.955–0.960) but weak against stock-flow and depensation (power 0.010–0.110)—an identification limit, not gate conservatism. Non-retention therefore reads as strong evidence where the instrument is powerful and weak evidence where it is not; reporting verdicts without operating characteristics conflates the two.

Keywords: model retention; out-of-sample forecast evaluation; reporting standard; operating characteristics; negative results; stock assessment; groundwater.